



Transfusion Policy Manual

Checking Procedure and Administration of Blood and Blood Products

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Group-Wide	All Clinical Staff who administer Blood products

Introduction

This document is designed as a general tool for clinical staff who administer blood and blood products. For product specific administration, refer to the relevant section.

Risk Statement

Non-compliance with this policy will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☐	Staff Safety	☐
Other:	☐		

Policy Principles

The SCGOPHCG Transfusion Policy Manual covers all aspects of transfusion from the decision to transfuse, consent, prescription, administration, as well as management of adverse transfusion events. The Transfusion Policy Manual (TPM) is updated and maintained by the SCGOPHCG Blood Management Committee.

This document is designed to direct staff in the safe administration of blood and blood products and should be utilised in conjunction with the [SCGOPHCG NPG Intravenous Therapy](#).

Procedure

ALL blood products **MUST** be prescribed and recorded on a SCGOPHCG Blood Product Transfusion Form, and a valid transfusion consent must be sighted.

- Observe the five moments for hand hygiene. Follow aseptic technique and use the risk assessment outlined in the Infection Prevention and Control manual. This must include use of appropriate Personal Protective Equipment (PPE) to prevent contamination and mucosal or conjunctival splash injuries.
- A transfusion **must not** be commenced without a valid transfusion consent form (unless it is an emergency) in accordance with the [NMHS Consent to treatment policy](#)

- Wherever possible, night transfusions in **clinically stable, non-bleeding** patients should be avoided between 2100 – 0700 (that is, transfusions should be completed by 2100hrs- staff commencing a transfusion should ensure that the transfusion can be completed prior to the night shift where nursing and medical staffing levels are minimal). This does not include the following:
 - a. Post-operative patients requiring transfusion.
 - b. Patients who are transferred from critical care areas with a transfusion in progress.
- Blood product transfusions must be commenced, or fractionated products confirmed for transfusion, within 30 minutes of removal from controlled storage. If the blood product cannot be commenced/confirmed within 30 minutes, then it must be returned to PathWest TMU (SCGH) or to the blood fridge (OPH) immediately.
- One patient compatibility label **MUST** remain on a fresh blood product.
- Rate of transfusion to be checked by 2 clinical staff at commencement of transfusion and at any required rate changes during transfusion.
- Inform the patient about possible adverse effects of transfusion and the importance of immediately reporting any new symptoms whilst blood product is being transfused (these may include but not limited to shivering, flushing, loin pain, pain in chest or extremities, rashes, abnormal bleeding, and shortness of breath).
- Ensure that the call bell is within easy reach of the patient.
- Provide the patient with written information about transfusion and its risks.
- The patient must be placed on a bed, trolley, or recliner chair during the transfusion episode.
- During the transfusion episode, the patient **must not leave the ward area** unless escorted.
- Patients receiving blood product therapy must be monitored closely for signs of adverse effects. This includes ongoing visual assessment as well as vital sign monitoring. Any adverse reactions should be dealt with immediately. Please refer to the [Care and Management of Transfusion Reactions](#) policy if this occurs.

1. Venous Access

Blood products must be transfused through the following:

1. A peripheral intravenous access device:

- 16-20 gauge is recommended for adults to reduce the risk of haemolysis. In paediatric patients and adults with fragile or difficult veins, a smaller gauge cannula can be used but may restrict the flow rate and impact on infusion times (avoid cubital fossa placement).
- In the critical bleeding or massive transfusion setting, large gauge IV access is recommended to achieve adequate flow rates to resuscitate the patient.

2. Central venous access devices and implanted ports

3. Peripherally inserted central catheter (PICC)

Note that PICC lines have smaller tubing diameters that can cause slower flow rates.

2. Transfusion Equipment

Red Blood Cells (packed cells), Fresh Frozen Plasma, Platelets and Cryoprecipitate **MUST** be administered through a blood giving IV set containing a 170-200micron filter to remove clots and debris that may have formed during storage.

- **Always** use a new intravenous administration set containing a 170-200micron filter for **platelet transfusions**. Cellular debris from previous transfusions would trap platelets in the filter of a used administration set.
- The route of administration must be identified on all administration lines as per [SCGOPHCG NPG Intravenous Therapy](#)
- Eight to ten red cell units may be transfused before the infusion set must be changed, depending on the quality of the flow rate. Otherwise, the infusion set must be changed post transfusion event, to a maximum 12 hours.
- Intravenous albumin and immunoglobulin may be transfused using a standard intravenous giving set via gravity or an electronic infusion pump.
- The intravenous line must be discarded after each transfusion episode.
- SCGH only: only one unit of red blood cells will be issued from TMU unless there is a need for rapid transfusion of large quantities of blood products. If multiple units are required, products should be issued from TMU in an esky or blood shipper with a specified return to TMU time visible.

3. Blood Warmers

Blood warmers are generally indicated for patients with clinically significant cold agglutinins (TMU will advise- refer to special requirements for blood products guideline) OR rapid infusion of blood products, that is, exceeding 50ml/kg/hr in adults.

- Red blood cells must only be warmed using a specifically designed commercial device with a visible thermometer and audible warning.
- The operating temperature must be recorded in the patient's infusion record.
- Improvised warming (e.g. putting the pack in hot water, in a microwave oven) **must never** be used. Such warming methods may damage red cells and cause harm to the patient.
- A limited number of blood warmers are available from the SCGH Hospital EquipmentStore or in Theatres at OPH.

4. Rapid Infusion Pump and External Pressure Bags

In some instances, blood products may be infused using external pressure devices. An external pressure bag, or the 'Level 1' rapid infusion pump may be used; however, the use of the 'Level 1' infusion pump is restricted to critical care areas.

- Establish that it is safe and appropriate for blood products to be administered using rapid infusion devices/external pressure bag prior to commencement.
- Large gauge venous access is required to quickly infuse blood products.
- Close observation should be maintained during transfusion of blood under pressure to avoid such possibilities as air embolism or extravasation.
- External pressure devices must not be used when transfusing platelets.
- **External pressure bag only:** Inflation pressure should never exceed 300mmHg and the pressure must be evenly distributed over the entire blood bag.

5. Compatible Concurrent Intravenous Fluids and Medications

- Normal Saline (0.9% NaCl) is the **only** intravenous fluid which can be added or used concurrently through the same tubing. No other medications or solutions should be added.

6. Preparation for administration before Blood Product is collected

<u>Preparation for Administration</u>
• Ensure you have the correct equipment, and there are enough staff available, and administration is being performed in a clinical area
• Ensure that there is a valid consent to Blood Products Form C, or Consent to Treatment Form A with YES ticked for consent to blood for surgical patients, and confirm that the patient understands the procedure. (Unless it is an emergency and/or the patient is unable to provide consent. In this case, NOK if contactable)
• IV access is patent. Minimum 22 Gauge cannula
• Patient ID band attached to patient and correct (Full name, DOB, UMRN & address)
• Observe and record baseline observations (within 1 hour of transfusion commencing¹, or immediately prior to commencing transfusion in a patient with active infection or is unstable): temperature (T), pulse (P), blood pressure (BP), respiration rate (RR), oxygen saturations (O₂ sats) and current oxygenation requirements. Document any pre-existing rashes.
• Administer Pre-Medication if prescribed

7. Collection of Blood Products

<u>Blood Product Collection</u>
Only collect the Blood Product when ready to start the transfusion
For SCGH: • Complete “Blood Product Transport Form” with correct patient ID (Full name, DOB, UMRN), product type and special requirements • Send the completed Blood Product Transport form to PathWest TMU (transfusion medicine unit) chute 92 • For emergency transfusions and plasma derived products in glass bottles such as albumin and intravenous immunoglobulin, send a HSA to collect from TMU • Blood shipper box: Keep the lid zipped closed and only remove blood products for transfusion as they are required. Return to TMU by time designated on shipper. For OPH: • Nurses and doctors can collect Blood Products from the Blood Bank refrigerator inside the theatre suite and complete the Blood Product Register or blood product pharmacy record on collection as per process in Appendix 1. <i>Only one clinical staff member (nurse or doctor) is required to collect blood products from blood bank refrigerator.</i>
• Check for any special requirements (such as irradiated blood products) on the blood product prescription.

8. Pre- Inspection of Blood Product

Inspection of Blood Product

Prior to the checking procedure and spiking of blood product, inspect the blood product for:

- Leakage and damage (especially at access ports)
- Clots and turbidity
- Discolouration

9. Checking Procedure of Blood Products

Double INDEPENDENT checking at the Patient's Bedside

1. Check the pt.'s ID with the transfusion order and compatibility label

2. Check that the pt.'s blood group and the donor blood group are the same OR compatible

3. Confirm that the correct type of blood product has been issued

4. Check that the 13 digit label # on the blood bag matches the compatibility label

5. Check the expiry date and time

**6. Are there any special transfusion requirements?
E.g. irradiated, pre-medication**

Prior to administration of blood or blood products **“double independent checking”** and **verification of the blood or blood product is required**. This requires two clinical staff members, one of whom must be a registered nurse or medical officer INDEPENDENTLY checking and verifying the blood and blood product.

Each staff member is accountable and responsible for the accuracy of the checking procedure and must complete all the checks independently of the other clinician. The checking procedure must occur immediately prior to the administration of each unit of blood product at the patient's bedside.

The staff member who spikes and hangs the blood product must be one of the two people who have undertaken the checking procedure.

CHECK: The patient or accompanying next of kin/legal guardian/carer must state their full name, date of birth (DOB) and residential address. Both staff must then confirm that these details are identical to the patient ID band.

Refer to [SCGOPHCG Patient Identification Policy](#) & [NMHS Patient Identification and Procedure Matching Policy](#)

CHECK: The patient details (first and family names, DOB & UMRN) match the patient details on the Blood Product Transfusion Form.

CHECK: The patient details on the Blood Product Transfusion Form match the patient details on the blood product compatibility label.

CHECK: The details on the PathWest compatibility label corresponds with the details on the blood pack:

- Correct blood product type
- ABO blood group compatibility
- 13-digit Blood Donation Number
- Expiry date and time
- Special requirements met (if any)

When the checking process has been completed and is deemed correct, affix the PathWest compatibility label onto the Blood Product Transfusion Form.

The two 'checkers' must sign the Blood Product Transfusion Form.
Document the date and time of commencement, and completion, of transfusion on the blood product transfusion form (MR826).

If there are any discrepancies with the bedside checking procedure, the blood product must not be transfused.

If there are any concerns or queries during checking procedure contact:

SCGH: PathWest TMU 6383 4018

OPH: PathWest 6457 8088. Out of hours contact on call scientist via A/H CNM

Blood Product Administration

Blood Product Administration

- Transfusion must be commenced by one of the staff who completed the checking process
- Commence transfusion within 30 minutes of removal from controlled storage
 - SCGH- time issued from TMU (refer to issue time on yellow label on blood product)
 - OPH- time removed from the theatre blood fridge
- The nurse must remain with the patient for AT LEAST the first 5 minutes post commencement of each blood product.
- **Infusion pump rates to be checked by 2 clinical staff members at the commencement of transfusion and at any rate changes**

Red Cell transfusions:

- commence at 75mls/hr for the first 15 minutes (excludes emergencies)
- providing there are no adverse reactions, at 15 minutes, increase to the prescribed rate

Observation requirements (T, P, BP, RR, O₂ sats, O₂ requirements) at:

- 15 minutes after commencement of each unit
- Hourly (from commencement of transfusion), and
- at completion.

More frequent vital signs monitoring may be required if patient is clinically unstable.

Complete within 3 hours of commencement of transfusion. Discard any remaining blood product and document reason for discard on medical record.

For **blood product administration** information of other blood products, refer to the relevant policy in section 7. Administration of blood and blood products in the [SCGOPHCG Transfusion Policy Manual](#)

Transfusion Reactions

- Monitor for signs of transfusion reaction as per the [Care and management of transfusion reactions](#) policy
- Follow the [Transfusion Reaction Algorithm](#)
- Immediate management- **STOP** the transfusion, assess patient (including vital signs), contact doctor to review patient, activate MET call if required.

The clinical outcome of the transfusion must be documented in the medical record.

Compliance, monitoring and evaluation

Non-adherence to this policy may result in adverse patient outcomes from incorrect checking and administration of blood products, and/or inadequate monitoring, identification and management of adverse reactions. The SCGOPHCG blood management committee is responsible for ensuring the effectiveness of this policy and monitors compliance and appropriate use of blood products through regular audit process.

Related Documents

Authorising Policy and Standard/s:

- NSQHS Standard 7 Blood Management

Standards, procedures, guidelines:

- [SCGOPHCG NPG Intravenous Therapy](#)
- [NMHS Patient Identification and Procedure Matching Policy](#)
- [SCGOPHCG Patient Identification Policy](#)
- Australian and New Zealand Society of Blood Transfusion and the Australian College of Nursing guidelines for the Administration of Blood Products
<https://anzsbt.org.au/wp-content/uploads/2019/10/ANZSBT-Administration-Guidelines-Revised-3rd-edition-Publication-Version-20191002.pdf>

Supporting documents

- Blood Product Transfusion Form

References

1. Australian and New Zealand society of blood transfusion, Australian College of Nursing. Guidelines for administration of blood products (Internet). Sydney: Australian & New Zealand Society of Blood Transfusion Ltd; 2024 Jan (cited 2024 Nov 5). Available from :<https://anzsbt.org.au/wp-content/uploads/2020/03/ANZSBT-Administration-Guidelines-Revised-3rd-edition-Publication-Version-FINAL-20191002.pdf>
2. The Joanna Briggs Institute. Medication Errors: Manual Independent Double Checking (Procedures) [Evidence Summary- Internet] Adelaide: Joanna Briggs Institute, 2022 [cited 2024 Nov 11]. Available from JBI
3. The Joanna Briggs Institute. Transfusion: Blood or Blood Product: [Recommended Practice - Internet]. Adelaide: Joanna Briggs Institute, 2021 [cited 2024 Nov 11]. Available from JBI.
4. The Joanna Briggs Institute. Transfusion: Blood Warmer [Recommended Practice - Internet]. Adelaide: Joanna Briggs Institute, 2022 [cited 2024 Nov 11]. Available from JBI.

Authorisation & Version Control

Policy Sponsor	Blood Management Committee Executive Sponsor				
Policy Contact	CNC Transfusion				
Date First Issued:	01/08/2021	Last Reviewed:	03/12/2024	Review Date:	05/11/2027
Version No.	3				
Approved by:	SCGOPHCG Blood Management Committee			Date:	10/12/2024
Endorsed by:	SCGOPHCG Blood Management Committee			Date:	10/12/2024
Standards Applicable:	NSQHS Standards (Version 2): 				

This document can be made available in alternative formats on request for a person with a disability.

APPENDIX 1

